

UNIT I : MATRICES AND SYSTEM OF LINEAR EQUATION

Matrices - Row Echelon form - Rank - Non homogeneous system of Linear equations - Homogeneous system of Linear Equations - Rouches Capelli Theorem (without proof) - Consistency - Gauss Jordan method - Gauss Seidel method.

(No direct 16 M) $\longrightarrow$ Confirm 8 mark

1 : Matrices And System of Linear Equations

1. Define Rank of a matrix. (2 M)

Let A be a $m \times n$, the rank of A is said to be r

if (i) There is atleast 1 minor of A of order r which does not vanish.

(ii) Every minor of A of order $(r+1)$ and higher order vanishes.

In other words. The rank of a matrix is the largest of the order of all the non vanishing minors of that matrix.

$\longrightarrow$ (3 M)
2. Define solution of system of Linear Equations

The set of values of $x_1, x_2, x_3, \dots, x_n$ which satisfies all the given m - Equations in the system simultaneously is called solution of system of Linear Equations.

3. Define consistent and Inconsistent of System of Linear equation. (2m)

If the system of linear equations $Ax=B$ has a solution ~~that~~ ^{then} is consistent otherwise the system is inconsistent.

4. State Rouché's Theorem. (2m)

Rouché's Theorem: [State necessary and sufficient condition of ^{system} linear non hom eqn is to be consistent]

The system of LE $Ax=B$ is consistent iff the coefficient matrix A and ^{aug} augmented matrix $[A, B]$ or $[A|B]$ are of the same rank
(i.e) $R[A] = R[A, B]$

5. What is the condition for the system of with n LEs to have (2m)

- (i) Exactly 1 solution.
- (ii) Infinitely many solution.
- (iii) No solution.

Sol.
(i) Given system of Equation can be written as $Ax=B$ then

(i) If $R[A] = R[A, B] = n$, then system of Equations are consistent and have unique solution.

(ii) If $R[A] = R[A, B] < n$, then S.O.E are consistent and have infinitely many solution.

(iii) If $R[A] \neq R[A, B]$, then system is inconsistent and it has no solution.

6. Test the consistency of the following S.O.E and solve if consistent. (8 m)

(i) $3x + y + z = 8$

$-x + y - 2z = -5$

$x + y + z = 6$

$-2x + 2y - 3z = -7$

The above eqns can be written as,

$$AX = B$$

(i.e)
$$\begin{bmatrix} 3 & 1 & 1 \\ -1 & 1 & -2 \\ 1 & 1 & 1 \\ -2 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 6 \\ -7 \end{bmatrix}$$

The augmented matrix is,

$$[A, B] = \left[\begin{array}{ccc|c} 3 & 1 & 1 & 8 \\ -1 & 1 & -2 & -5 \\ 1 & 1 & 1 & 6 \\ -2 & 2 & -3 & -7 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ -1 & 1 & -2 & -5 \\ 3 & 1 & 1 & 8 \\ -2 & 2 & -3 & -7 \end{array} \right]$$

$$R_1 \leftrightarrow R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & -2 & -2 & -10 \\ 0 & 4 & -1 & 5 \end{array} \right]$$

$$R_2 \rightarrow R_2 + R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_4 \rightarrow R_4 + 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2$$

$$R_4 \rightarrow R_4 - 2R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_4 \rightarrow 3R_4 + R_3$$

$$\longrightarrow (A)$$

It is an echelon form

It has 3 non zero rows

$$\therefore R[A] = R[A, B] = R = n = 3$$

$\Rightarrow$ The system of equation is consistent
& it has unique solution.

$$(A) \Rightarrow R_1 \Rightarrow x + y + z = 6$$

$$R_2 \Rightarrow 2y - z = 1$$

$$R_3 \Rightarrow -3z = -9$$

$$\boxed{z = 3}$$

$$2y - 3 = 1$$

$$2y = 4$$

$$\boxed{y = 2}$$

$$x + 2 + 3 = 6$$

$$x + 5 = 6$$

$$\boxed{x = 1}$$

Solutions: $x = 1$; $y = 2$; $z = 3$ //

$$(ii) \quad 2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$x_1 - x_2 + x_3 + 2x_4 = 6$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

The equation can be written as,

$$[A, B] = \left[\begin{array}{cccc|c} 2 & 1 & 2 & 1 & 6 \\ 1 & -1 & 1 & 2 & 6 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right], \text{ Augmented matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 2 & 1 & 2 & 1 & 6 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 3 & 0 & -3 & -6 \\ 0 & 7 & -1 & -11 & -25 \\ 0 & 4 & -3 & -3 & -2 \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 3 & 0 & -1 & -2 \\ 0 & 7 & -1 & -11 & -25 \\ 0 & 4 & -3 & -3 & -2 \end{array} \right] \quad R_2 \rightarrow R_2/3$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -1 & -4 & -11 \\ 0 & 0 & -3 & 1 & 6 \end{array} \right] \quad \begin{array}{l} R_3 \rightarrow R_3 - 7R_2 \\ R_4 \rightarrow R_4 - 4R_2 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -1 & -4 & -11 \\ 0 & 0 & 0 & 13 & 39 \end{array} \right] \xrightarrow{R_4 \rightarrow R_4 - 3R_3} (A)$$

It is in Echelon form.

It has four non zero rows.

$$\therefore R[A] = R[A, B] = n = 4$$

$\Rightarrow$ The system is consistent and it has unique solution.

$$(A) \Rightarrow R_1 = x_1 - x_2 + x_3 + 2x_4 = 6$$

$$R_2 = x_2 - x_4 = -2$$

$$R_3 = -x_3 - 4x_4 = -11$$

$$R_4 = 13x_4 = 39$$

$$\boxed{x_4 = 3}$$

$$\boxed{x_2 = 1}$$

$$-x_3 - 4(3) = -11$$

$$-x_3 - 12 = -11$$

$$-x_3 = 1$$

$$\boxed{x_3 = -1}$$

$$x_1 - 1 + (-1) + 2(3) = 6$$

$$x_1 - 2 + 6 = 6$$

$$\boxed{x_1 = 2}$$

Solutions: $x_1 = 2$
 $x_2 = 1$
 $x_3 = -1$

$$\begin{aligned}
 \text{(iii)} \quad & 2x + y + 5z + w = 5 \\
 & x + y + 3z - 4w = -1 \\
 & 3x + 6y - 2z + w = 8 \\
 & 2x + 2y + 2z - 3w = 2
 \end{aligned}$$

$$\begin{aligned}
 \text{7. (i)} \quad & x - 4y - 3z + 16 = 0 \\
 \text{(8m)} \quad & 2x + 7y + 12z = 48 \\
 & 4x - y + 6z = 16 \\
 & 5x - 5y + 3z = 0
 \end{aligned}$$

Sol

$$\left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 2 & 7 & 12 & 48 \\ 4 & -1 & 6 & 16 \\ 5 & -5 & 3 & 0 \end{array} \right]$$

Augmented matrix

$$\sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 15 & 18 & 80 \\ 0 & 15 & 18 & 80 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \\ R_4 \rightarrow R_4 - 5R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array} \text{--- (A)}$$

It is in Echelon form,

It has 2 non zero rows.

$$\therefore R[A] = R[A|B] = 2 < n = 3$$

$\Rightarrow$ The System of Equation is Consistent
and has infinitely many solutions.

$$(A) \Rightarrow R_1 = x - 4y - 3z = -16 \text{ ————— (1)}$$

$$R_2 = 15y + 18z = 180 \text{ ————— (2)}$$

Let $z = k \in \mathbb{R}$ in (2),

$$(2) \Rightarrow 15y + 18k = 180$$

$$15y = 180 - 18k$$

$$y = \frac{180 - 18k}{15}$$

$$y = \frac{180}{15} - \frac{18}{15}k$$

$$\boxed{y = \frac{16}{3} - \frac{6}{5}k}$$

$$(1) \Rightarrow x - 4\left[\frac{16}{3} - \frac{6}{5}k\right] - 3k = -16$$

$$x - \frac{64}{3} + \frac{24k}{5} - 3k = -16$$

$$x + \frac{24k}{5} - 3k = -16 + \frac{64}{3}$$

$$x + \frac{9k}{5} = \frac{-48 + 64}{3}$$

$$\boxed{x = \frac{16}{3} - \frac{9k}{5}}$$

$$\boxed{z = k}$$

$$(ii) \quad x_1 + 2x_2 - x_3 - 5x_4 = 4 \quad \text{--- (A)}$$

$$x_1 + 3x_2 - 2x_3 - 7x_4 = 5$$

$$2x_1 - x_2 + 3x_3 = 3$$

$$\begin{bmatrix} 1 & 2 & -1 & -5 & | & 4 \\ 1 & 3 & -2 & -7 & | & 5 \\ 2 & -1 & 3 & 0 & | & 3 \end{bmatrix} = [A | B]$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & -5 & | & 4 \\ 0 & 1 & -1 & -2 & | & 1 \\ 0 & -5 & 5 & 10 & | & -5 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & -5 & | & 4 \\ 0 & 1 & -1 & -2 & | & 1 \\ 0 & -5 & 5 & 10 & | & -5 \end{bmatrix} \quad R_3 \rightarrow R_3 / 5$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & -5 & | & 4 \\ 0 & 1 & -1 & -2 & | & 1 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 + R_2 \\ \text{--- (A)} \end{array}$$

It is in Echelon form.

It has 2 non zero rows.

$$\therefore R[A] = R[A | B] = 2 < n = 4$$

$\Rightarrow$ The S.O.E is consistent and has infinitely many solutions.

$$(A) \Rightarrow R_1 = x_1 + 2x_2 - x_3 - 5x_4 = 4 \quad \text{---(1)}$$

$$R_2 = x_2 - x_3 - 2x_4 = 1 \quad \text{---(2)}$$

Let $\boxed{x_3 = k_1}$ $\boxed{x_4 = k_2}$ for $k \in \mathbb{R}$.

$$(2) \Rightarrow x_2 - k_1 - 2k_2 = 1$$

$$\boxed{x_2 = 1 + k_1 + 2k_2}$$

$$(1) \Rightarrow x_1 + 2(1 + k_1 + 2k_2) - \overset{k_1}{x_3} - 5k_2 = 4$$

$$x_1 + 2 + 2k_1 + 4k_2 - k_1 - 5k_2 = 4$$

$$x_1 + k_1 - k_2 = 4 - 2$$

$$\boxed{x_1 = 2 - k_1 + k_2}$$

8. Test the Consistency of following SDE:-

$$(i) \quad x - 2y - 3z = 2$$

$$3x - 2y = -1$$

$$-2y - 3z = 2$$

$$y + 2z = 1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 3 & -2 & 0 & -1 \\ +0 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$8. (i) \quad x - 2y - 3z = 2$$

$$3x - 2y = -1$$

$$-2y - 3z = 2$$

$$y + 2z = 1$$

Augmented matrix is

$$[A, B] = \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 3 & -2 & 0 & -1 \\ 0 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{array} \right] \quad R_2 \rightarrow R_2 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & -1 & 11 \end{array} \right] \quad \begin{array}{l} R_3 \rightarrow 2R_3 + R_2 \\ R_4 \rightarrow 4R_4 - R_2 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 30 \end{array} \right] \quad R_4 \rightarrow 3R_4 + R_3$$

It is in Echelon form.

$$\therefore R[A] = 3 \neq R[A, B] = 4$$

$\therefore$ The SDE is inconsistent & it has no solution

8. (ii) $x_1 + x_2 + x_3 = 1$

$$2x_1 - 2x_2 + 3x_3 = 1$$

$$x_1 - x_2 + 2x_3 = 5$$

$$x_1 + x_2 + x_3 = 2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & -2 & 3 & 1 \\ 1 & -1 & 2 & 5 \\ 1 & 1 & 1 & 2 \end{array} \right]$$

9. (i) Find the values of k for which the equations

$$x + y + z = 1$$

$$x + 2y + 4z = k$$

$$x + 4y + 10z = k^2 \quad \text{may have solution.}$$

For these values of k find the solution also.

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & k \\ 1 & 4 & 10 & k^2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 3 & 9 & k^2-1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 0 & 0 & k^2-3k+2 \end{array} \right] \quad R_3 \rightarrow R_3 - 3R_2$$

$$\begin{aligned} k^2 - 1 - 3(k-1) \\ k^2 - 1 - 3k + 3 \\ k^2 - 3k + 2 \end{aligned}$$

$\sim$ It is in echelon form.

$$\nabla \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 0 & 0 & (k-1)(k-2) \end{array} \right] \text{---(A)}$$

Here $R[A] = 2 = R[A, B]$

$$(k-1)(k-2) = 0$$

$k=1$
$k=2$

i) Put $k=1$ in (A) $\Rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x+y+z=1$$

$$y+3z=0$$

$$x-3t+t=1$$

$$\boxed{x = 1 + 2t}$$

Put $z=t \in \mathbb{R}$

$$\boxed{y = -3t}$$

$$\boxed{z = t}$$

$$\Rightarrow t \in \mathbb{R}$$

(ii) Put $K=2$ in (A) $\Rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x + y + z = 1$$

$$y + 3z = 1$$

$$\boxed{z = \lambda}$$

$$\boxed{y = 1 - 3\lambda}$$

$$\boxed{y = 1 - 3\lambda}$$

$$x + (1 - 3\lambda) + \lambda = 1$$

$$x + 1 - 3\lambda + \lambda = 1$$

$$x + 1 - 2\lambda = 1$$

$$\boxed{x = 2\lambda}$$

$$\forall \lambda \in \mathbb{R}$$